

[Primary congenital glaucoma].

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Primary congenital glaucoma (PCG) is a rare genetic disease usually diagnosed during the first year of life. It occurs because of developmental anomalies of the chamber angle that prevents drainage of aqueous humor, thereby elevating intraocular pressure. Its incidence is 1 in 10,000 live newborns in Western societies and 1 in 1,200 live newborns in the Arab-Bedouin population of the Negev region in Israel. Most cases of PCG appear to be sporadic. The cytochrome P450B1 gene located within the GLC3A locus on chromosome 2p21 is mutated in individuals with PCG. The triad of epiphora, photophobia, and blepharospasm is classical for PCG. General anesthesia is usually required for an adequate examination of intraocular pressure, corneal diameter, optic disc, and axial length in young children. Congenital glaucoma is almost always managed surgically, with medical therapy being used only as a temporizing measure before surgery or when surgical intervention has repeatedly failed. At least 50% of eyes with PCG presenting at birth will become legally blind (visual acuity < 6/60). Patients with PCG require follow-up examinations throughout their lives.

Congenital glaucoma and neurofibromatosis in a monozygotic twin: case report and review of the literature.

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We describe a dramatic case of an identical twin presenting at birth with unilateral congenital glaucoma. Because of the suspicion of neurofibromatosis 1 a magnetic resonance image of the neural axis was obtained, which revealed a plexiform neurofibroma with spinal cord impingement. Diagnosis of neurofibromatosis 1 was confirmed by 3 months of age with the emergence of café-au-lait spots. This case was compared with all 19 reports published in the English literature of neurofibromatosis 1 associated with congenital glaucoma. Initial presentation, family history, characteristics of the clinical syndrome, and outcome of glaucoma in infants with neurofibromatosis 1 and congenital glaucoma were reviewed. A plexiform neurofibroma of the ipsilateral eyelid was present in eight patients and ipsilateral facial hypertrophy occurred in three patients. Café-au-lait spots appeared between the ages of 5 weeks and 8 years; none of the patients were reported to have café-au-lait spots at birth. Newborns with unilateral congenital glaucoma should raise high suspicion for neurofibromatosis 1 and its associated findings, which might need immediate intervention.

Screening of West Siberian patients with primary congenital glaucoma for CYP1B1 gene mutations.

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Primary congenital glaucoma (PCG) is a visual organ pathology that leads to progressive blindness and poor vision in children. Its main cause is an anomaly of the anterior chamber angle. Most cases of PCG are sporadic, but familial cases with an autosomal recessive (predominantly) and autosomal dominant (rare) type of inheritance have been described. Congenital glaucoma is a rare condition (1 case per 10,000-20,000 newborns), but its prevalence is substantially higher (up to 1 case per 250 newborns) in countries where consanguineous marriages are common. Mutations in the CYP1B1 gene, which encodes

cytochrome P450 1B1, are the most common cause of autosomal recessive primary congenital glaucoma. This enzyme is known to be involved in retinoic acid metabolism and is necessary for normal eye development. The aim of this work was to assess the polymorphism of the CYP1B1 gene among West Siberian patients with primary congenital glaucoma. Direct automatic Sanger sequencing of exons and adjacent splicing sites of the CYP1B1 gene was carried out in 28 people with the PCG phenotype from a West Siberian region. As a result, in the sample of the white population we examined, pathogenic variants previously described in other ethnic groups were revealed: E387K (rs55989760), R444* (rs377049098), R444Q (rs72549376), and P437L (rs56175199), as well as novel single-nucleotide deletion p.F114Lfs*38 in the CYP1B1 gene. The latter can cause a frame shift, changed amino acid composition, and a formation of truncated in the protein. None of the detected mutations were found in the control sample of ophthalmologically examined individuals without PCG (100 people). Variants R444* (rs377049098) and R444Q (rs72549376) were not found in the general population sample either (576 randomly selected West Siberia residents). All the detected mutations caused the development of the autosomal recessive form of primary congenital glaucoma. The most severe clinical phenotype was observed in carriers of mutations in codon 444 of the gene. Consequently, in children with signs of increased intraocular pressure, molecular genetic analysis of the CYP1B1 gene is advisable for early diagnosis and timely initiation of PCG therapy.

An Insight into Primary Congenital Glaucoma.

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Glaucoma is the second most prominent cause of impaired vision in the world. Over 60 million individuals are presently affected, and 12 million are sightless as a result. Primary congenital glaucoma (PCG) is a childhood disease that can lead to blindness in newborns and very young children. The rate of occurrence of PCG varies in different communities and across geographical boundaries, and its etiology is unknown. It is caused by genetic structural defects in the trabecular meshwork and makes its appearance in newborns and children no older than three years. PCG is most prevalent in populations with high rates of consanguineous marriages. It is categorized by inappropriate development of the eye's aqueous outflow system, causing increased intraocular pressure (IOP) and leading to swelling of the cornea, epiphora, discomfort or pain, enlargement of the eyeball (buphthalmos), corneal opacity, and optic nerve damage. PCG is classified as an autosomal recessive disorder involving four loci. The main culprit is CYP1B1, at locus GLC3A. PCG is also linked with loci GLC3B and GLC3C; however, their genetic factors have only recently been recognized. The gene LTPB2 at locus GLC3D, plays an important role in tissue healing and cell attachment. Trabeculectomy and gonioscopy are effective treatments for PCG. Additional efforts are essential to provide timely screening of children and, most important, to assign sufficient resources to allow healthcare workers to reduce the rate of avoidable blindness in developing countries.

Congenital glaucoma.

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Congenital and juvenile glaucoma are associated with goniodysgenesis and currently thought to be the result of neural crest cell abnormal terminal induction or migration. Infantile glaucoma may be primary, or may be associated with syndromes such as Sturge-Weber, Rieger's and others. Differentiation from other childhood causes of cloudy cornea such as endothelial dystrophies is essential. Evaluations under

anesthesia are often needed and present their own problems in diagnosis.

Newborn primary congenital glaucoma: 2005 update.

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Newborn primary congenital glaucoma is an unusual and important subtype of primary congenital glaucoma. Affected patients typically manifest specific signs that facilitate its recognition at birth and become important in the assessment and determination of appropriate treatment for this severe expression of primary congenital glaucoma. The medical records of 35 patients with newborn primary congenital glaucoma were reviewed to determine its clinical manifestations and response to therapy. The published literature related to newborn primary congenital glaucoma was reviewed and referenced. Newborn primary congenital glaucoma is recognized at birth because of the associated corneal opacification. Ocular examination reveals anterior segment abnormalities of the cornea, iris, and filtration angle as well as related elevated intraocular pressure. Genetic analysis of a subset of patients with newborn primary congenital glaucoma confirmed its relationship with the less severe infantile form of primary congenital glaucoma, which is characterized by favorable results after goniosurgery. In contrast, goniosurgery was found to have unfavorable results and be less successful compared with trabeculectomy and glaucoma drainage tube shunts as initial therapy for newborn primary congenital glaucoma. Newborn primary congenital glaucoma is an important subtype of primary congenital glaucoma. It can be differentiated from the more frequent and familiar infantile type by history and careful anterior segment examinations to enable and support appropriate choices for its successful surgical treatment.

Newborn Glaucoma: A Neglected Manifestation of Congenital Rubella Syndrome.

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To study the frequency of intrauterine rubella infection in a cohort of neonatal-onset glaucoma (NOG) and its effect on the treatment outcomes. Prospective cohort study. Infants with NOG presenting to the pediatric glaucoma service at a tertiary care center in northern India at Chandigarh between January 1, and September 30, 2018, with a minimum postoperative follow-up of 1 year were included in this study. Rubella immunoglobulin-M (IgM) antibodies were tested in all patients. Surgery for intraocular pressure (IOP) control comprised combined trabeculotomy with trabeculectomy or goniotomy. Presenting features and treatment outcomes at 1 year were compared among infants with and without rubella-IgM antibodies. Of 27 eligible infants, 7 (25.9%) were rubella-IgM positive, and all had bilateral glaucoma. One eye in a rubella-positive baby was an unsightly blind eye and required a prosthesis. A total of 13 eyes of 7 patients were compared with 34 eyes of 20 rubella-negative patients. There was no significant difference in mean age and IOP at presentation. Rubella-positive infants had significantly more opaque corneas ($P < 0.001$), shorter eyes ($P < 0.001$), and smaller corneal diameters ($P = 0.007$) at presentation. Two patients in the rubella group had bilateral cataracts, 3 patients had heart disease, and 1 patient had sensorineural deafness. Salt-and-pepper retinopathy was noted in both rubella-positive infants where fundus evaluation was possible. Despite comparable IOP control ($P = 0.51$), the corneal clarity remained significantly worse in the rubella-positive patients ($P = 0.02$). Twenty-five percent of those with newborn glaucoma had underlying intrauterine rubella infection, thus making them susceptible for development of congenital rubella syndrome (CRS). Neonatal-onset glaucoma is an important component of CRS, which

may present without buphthalmos and persistent corneal clouding despite good IOP control.

Molecular characterization of newborn glaucoma including a distinct aniridic phenotype.

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To characterize the underlying genetic defect in otherwise healthy Saudi newborns with buphthalmos, including those with iris abnormalities. Prospective case series of affected Saudi Arabian probands who were referred for genetic counseling over a 4 year period. All had CYP1B1 sequencing. Selected patients with visible iris abnormalities had PAX6, FOXC1, and PITX2 sequencing. CYP1B1-negative patients had LTBP2 sequencing. All 67 probands had corneal enlargement with variable haze/scarring evident to caregivers at birth; 46 had a family history of infantile or early childhood glaucoma. All families were consanguineous except for 6, 2 of which were endogamous. Eight probands had mild ectropion uveae with partial aniridia; 2 probands had thick scarred corneas that precluded careful iris examination. Homozygous or compound heterozygous CYP1B1 mutations were identified in 91% (61/67), including all 8 probands with ectropion uveae and partial aniridia. The common Saudi mutation p.G61E occurred in most cases (38 homozygous, 8 compound heterozygous). Four novel mutations were identified (p.N252K, p.V460E, p.S485F, p.N519D). No mutations were identified in the other screened genes. Newborn glaucoma on the Arabian Peninsula is typically CYP1B1-related even in the setting of developmental iris abnormality. Mild iris ectropion with partial aniridia in a newborn with glaucoma suggests mutations in CYP1B1 rather than in other genes associated with anterior segment dysgenesis. On the Arabian Peninsula p.G61E mutations are the major cause of newborn glaucoma but novel CYP1B1 mutations continue to be documented. The fact that the 9% of cases that were CYP1B1-negative did not have mutations in LTBP2 suggests that there exists at least 1 additional locus for this condition.

Neonatal-Onset Congenital Ectropion Uveae: A Distinct Phenotype of Newborn Glaucoma.

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To describe neonatal-onset congenital ectropion uveae (N-CEU) as a distinct clinical entity of newborn glaucoma (NG) and to study its significance toward the severity and outcome of NG. Prospective clinical cohort study. The study took place at a tertiary care postgraduate teaching institute. It included consecutive patients with NG who presented between July 1, 2016 and September 30, 2017, with a minimum postoperative follow-up of 1 year. Infants with any ocular anomaly apart from CEU were excluded. Patients with N-CEU were compared with those with neonatal-onset primary congenital glaucoma (N-PCG). All infants underwent goniotomy or trabeculotomy, with trabeculectomy depending on corneal clarity. Clinical features at presentation and outcome 1 year after surgery were defined as good or satisfactory if intraocular pressure was ≤ 16.0 mm Hg under anesthesia without or with topical medications, respectively, and poor if the infant required additional surgery. Twenty eyes of 10 patients with N-CEU were compared with 16 eyes of 9 patients with N-PCG. Infants with N-CEU had significantly worse corneal clarity (mean grade 2.0 ± 0.7 vs 1.4 ± 0.8 ; $P = .026$) and poorer outcomes compared with those with N-PCG. Seven of 16 (43.7%) eyes with N-PCG had a cornea clear enough at presentation for a goniotomy compared with only 2 of the 20 (10%) eyes with N-CEU ($P = .026$). Thirteen of 16 (81.2%) eyes with N-PCG had a good or satisfactory outcome compared with 6 of 20 (30%) eyes with N-CEU ($P =$

.001). N-CEU appears to be distinct from the unilateral CEU in older patients described in the literature and may be considered a poorer prognosis phenotype of neonatal-onset glaucoma.

Newborn Glaucoma: Do not Forget Infections.

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Intrauterine infections can affect various structures of the developing fetal eye. Rubella infection results in congenital cataracts, keratopathy, retinopathy and less commonly, glaucoma. Ophthalmic manifestations of intrauterine cytomegalovirus (CMV) infection have been reported to be chorioretinitis, optic nerve colobomas, and corneal opacities, but have not been implicated in congenital cataract or congenital glaucoma. Concurrent infection with both rubella and CMV virus has not been reported. We report concurrent rubella and CMV infection in a baby born with corneal opacification, severe congenital glaucoma, and congenital cataract. It is important to recognize these babies early and investigate for intrauterine infections rather than assume they are all primary congenital glaucoma. Involvement of the cornea, glaucoma, and cataract make management of these babies a major challenge requiring a multidisciplinary team approach.
